Chapter 1

Basic Circuit Elements and Fundamental Laws

1-1 Electrical Energy and Voltage

The amount of energy that is required to move a charge of Q coulomb from one point to another point against the electric field intensity is known as electrical energy. Constant electrical energy is denoted by W and time varying electrical energy is represented by w. The unit of electrical energy is watt second. Figure 1.1 explains electrical energy.

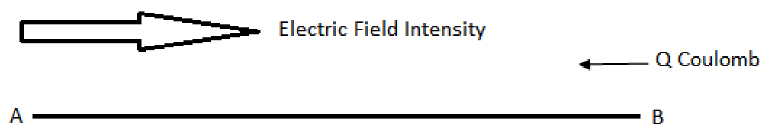

Figure 1.1: Electrical Energy

Electric field intensity is the force per unit positive charge in volts per meter and it is denoted by E. If d defines the distance between points A and B , then electrical energy can be calculated as

$$W = Fd \tag{1.1}$$

$$W = QEd \tag{1.2}$$

The amount of energy that is required to move a unit positive charge from one point to another point against the electric field intensity is known as voltage or potential difference between the two points. Time Constant voltage is denoted by V and time varying voltage is represented by v. The unit of voltage is volt. Figure 1.2 explains voltage between points A and B

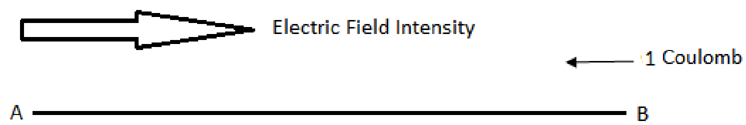

Figure 1.2: Voltage or Potential Difference

The voltage between points A and B is given by

$$V = \frac{W}{Q} \tag{1.3}$$

Therefore

$$V = Ed \tag{1.4}$$

The rate of motion of charge in a conductor defines current. Time constant current is represented by I and the time varying current is denoted by i. Current is given by equation 1.5.

$$I = \frac{Q}{t} \tag{1.5}$$

We know that energy per unit time is known as power. Time constant power is represented by P and the time varying power is denoted by p. The unit of power is watt and power is given by equation 1.6.

$$P = \frac{W}{t} \tag{1.6}$$

$$P = \frac{QV}{t} \tag{1.7}$$

Therefore electrical power can be calculated with the help of voltage and current

$$P = VI \tag{1.8}$$

D 1.1: A 12V battery is charged for 4 hours with a current of 2A. 60% of the energy is stored as chemical energy and the remaining energy is lost. If electricity costs Rs 0.02 per watt-second, then determine the cost of charging the battery, the amount of energy that is stored as chemical energy and the energy that is lost.

Solution:

$$V = 12V$$

$$I = 2A$$

$$t = 4 \times 3600 = 14400 \text{ seconds}$$

$$P = VI = 24 \text{ Watts}$$

$$W = Pt = 345600 \text{ Watt Second}$$

$$Cost = 345600 \times 0.02 = Rs\ 6912$$

The amount of energy stored as chemical energy $= 345600 \times 0.6 = 207360\ Jouls$

The amount of energy lost $= 345600 - 207360 = 138240$ watt-second.

1-2 Resistor and Ohm's Law

Resistor is a passive circuit element, if it is connected across a voltage source (active circuit element), it will take energy from the source. A resistor is connected across a variable voltage source as shown in Figure 1.3.

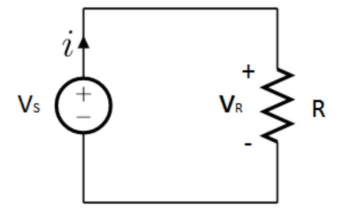

Figure 1.3: Circuit Diagram for Ohm's Law

Ohm's law states that *the voltage across a resistor is directly proportional to the current in the resistor provided the resistance of the resistor is held constant*.
Mathematically

$$v_R \propto i$$

By increasing voltage across the resistor, the current increases linearly as demonstrated in Figure 1.4.

$$v_R = iR \tag{1.9}$$

Where R is the constant of proportionality in equation 1.9. The resistance of a conductor depends upon the material, length and cross sectional area of the conductor and is given by

$$R = \frac{\rho \ell}{A} \tag{1.10}$$

Where ρ in equation 1.10 represents resistivity of the material of the conductor, ℓ represents length and A represents cross-sectional area. The current i in the above mentioned voltage source flows from the negative terminal of the voltage towards the positive terminal. This type of voltage is known as voltage rise. While the same current flows from the positive polarity of the voltage v_R towards the negative polarity, so this type of voltage is known as voltage drop. Kirchhoff's voltage law states that *sum of the voltage rises in a loop is always equal to sum of the voltage drops.*

So

$$v_s = v_R$$

The graphical representation of ohm's law is given in Figure 1.4. The relationship between voltage and current is a straight line passing through the origin of the two coordinates. This type of relationship is called linear relationship. Any circuit element

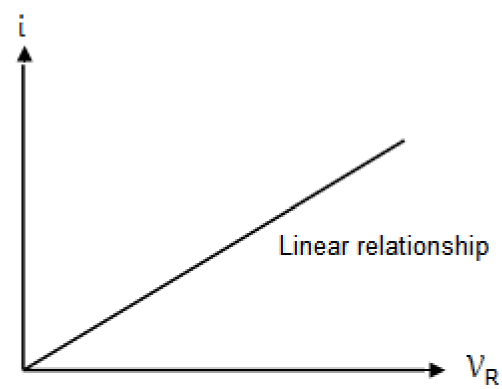

Figure 1.4: Graphical Representation of Ohm's Law

that has a linear relationship between voltage and current is known as linear circuit element. Resistor is a linear circuit element and other examples are inductor and capacitor. According to the law of conservation of energy the power supplied by the voltage source in Figure 1.3 will be equal to the power consumed by the resistor. The power consumed by the resistor is converted to heat energy that is dissipated in the air. The time varying power consumed by the resistor can be calculated with the help of any one the following three equations

$$p_s = p_R = v_R \times i \tag{1.11}$$

$$p_R = i^2 R \tag{1.12}$$

$$p_R = \frac{v_R^2}{R} \tag{1.13}$$

D 1.2: Let the voltage across the source in the circuit diagram of Figure 1.3 is 10V and resistance of the resistor is 5Ω. Determine the current in the resistor and power consumed by it.

Solution:

$$i = \frac{v_s}{R} = \frac{10}{5} = 2A$$

$$p_R = i^2 R = 4 \times 5 = 20W$$

1-3 Capacitor

It is a passive circuit element that stores electrical energy in its electric field and this is why it is known as energy storing device. It consists of two metallic plates having area of A m^2. A dielectric material with dielectric constant of ε_r is placed between the two plates. The distance between the two plates of the capacitor is denoted by d . The symbol of a capacitor is shown in Figure 1.5.

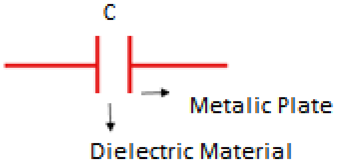

Figure 1.5: Symbol of Capacitor

The capacitance of a capacitor can be varied by varying any one the three parameters in equation 1.14.

$$C = \frac{A\varepsilon}{d} \tag{1.14}$$

As mentioned earlier, A is the area of the metallic plate, ε is the permittivity of the dielectric material and d represents separation between the plates. A time varying voltage is applied across the capacitor as shown in Figure 1.6. Charge starts accumulating on the plates of the capacitor and it depends on the applied voltage.

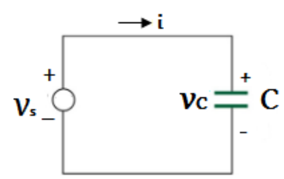

Figure 1.6: Capacitor in a Circuit

By increasing voltage across the capacitor charge on the plates will increase linearly and vice versa.

$$q \propto v_c$$

$$q = Cv_c \tag{1.15}$$

Where C is the constant of proportionality. Differentiating both sides of the above equation with respect to time, we obtain

$$\frac{dq}{dt} = C\frac{dv_c}{dt} \tag{1.16}$$

As $\frac{dq}{dt}$ represents the time varying current in the capacitor, Therefore

$$i = C\frac{dv_c}{dt} \tag{1.17}$$

The above equation reveals that if we apply dc voltage source across a capacitor it will block the dc current. In other words capacitor behaves like an open circuit for dc voltage. The differential voltage across a capacitor can be determined using following equation.

$$dv_c = \frac{1}{C}idt \tag{1.18}$$

Integrating both sides of equation 1.18, we obtain the time varying voltage across the capacitor

$$v_c = \frac{1}{C}\int idt \tag{1.19}$$

According to KVL, the only voltage rise v_s in the loop will be equal to the only voltage drop v_c that is

$$v_s = v_c \tag{1.20}$$

According to the law of conservation of energy power supplied by the voltage source in Figure 1.6 will be equal to the power taken by the capacitor.

$$p_s = p_c = v_c \times i \tag{1.21}$$

Putting the value of i in equation 1.21, we get the following equation.

$$p_c = C v_c \frac{dv_c}{dt} \tag{1.22}$$

The differential energy that is stored in the electric field of this capacitor is given by

$$dw = p_c \times dt$$

$$dw = C v_c \, dv_c \tag{1.23}$$

The integration on both sides gives the total energy that is stored in the electric field of this capacitor.

$$\int dw = C \int v_c \, dv_c$$

$$w = \frac{1}{2} C v_c^2 \tag{1.24}$$

1-4 Inductor and Faraday's Law

Inductor is basically a coil of N turns as shown in Figure 1.7. A simple straight conductor also behaves like inductor but its inductive effect is very small as compared to a coil having N turns.

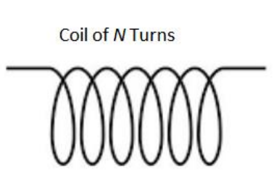

Figure 1.7: Symbol of Inductor

A time varying voltage is applied across the inductor as shown in Figure 1.8. Current starts flowing and this current generates time varying magnetic flux of Ø weber in the vicinity of the inductor. This time varying magnetic flux can be calculated with the help of following equation.

$$\emptyset = \frac{L}{N} i \tag{1.25}$$

Inductance L is given by

$$L = \frac{N\emptyset}{i} \tag{1.26}$$

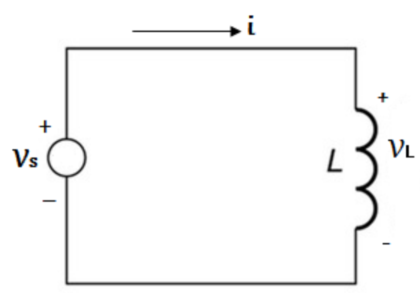

Figure 1.8: Inductor in a Circuit

Equation 1.26 reveals that a simple straight conductor also behaves like an inductor and its inductance can be calculated as

$$L = \frac{\emptyset}{i} \tag{1.27}$$

Differentiating both sides of equation 1.25 with respect to time, we get

$$N\frac{d\emptyset}{dt} = L\frac{di}{dt} \tag{1.28}$$

The inductor is located in its own time varying magnetic field and variation in the strength of the magnetic field will induce some voltage across this inductor, which is given by the mathematical model of Faraday's Law

$$v_L = \frac{Nd\emptyset}{dt}$$

This is the time varying voltage across the inductor which can be determined with the help of following equation as well

$$v_L = \frac{Ldi}{dt} \tag{1.29}$$

The above equation reveals that if we apply a DC voltage across an inductor, then it will behave like an ideal conductor (short circuit) as the voltage across this inductor will be zero. The differential current in an inductor can be determined using following equation.

$$di = \frac{1}{L} v_L dt$$

Integrating both sides of the above equation, we obtain the time varying current in the inductor.

$$i = \frac{1}{L} \int v_L dt \tag{1.30}$$

According to KVL, the only voltage rise v_s in the loop will be equal to the only voltage drop v_L , that is

$$v_s = v_L \tag{1.31}$$

According to the law of conservation of energy power supplied by the voltage source in Figure 1.8 will be equal to the power taken by the inductor.

$$p_s = p_L = v_L \times i \tag{1.32}$$

Putting the value of v_L in equation 1.32, we obtain

$$p_L = Li \frac{di}{dt} \tag{1.33}$$

The differential energy that is stored in the magnetic field of this inductor is given by

$$dw = p_L \times dt \tag{1.34}$$

The integration on both sides gives the total energy that is stored in the magnetic field of this inductor

$$\int dw = L \int i \, di$$

$$w = \frac{1}{2} Li^2 \tag{1.35}$$

D 1.3: Consider the sketch for v_c as shown in Figure 1.9. Sketch i and p_C as a function of time. Capacitance of the capacitor is 10F.

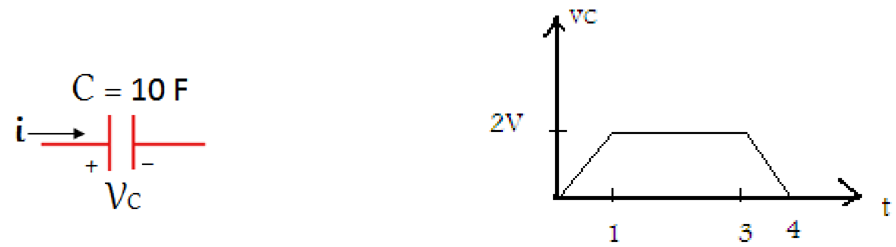

Figure 1.9: Capacitor and Sketch for the Voltage across the Capacitor

Solution:

We divide the graph into three regions.

Region # 1: $(0 \leq t \leq 1)$

In this region v_c is a straight line passing through the origin. Equation of this straight line is

$$v_c = mt + c$$

Where $m = 2$ is the slope of this line and as the line is passing through the origin therefore $c = 0$.
So

$$v_c = 2t$$

Region # 2: $(1 \leq t \leq 3)$

The voltage in this region is constant, therefore

$$v_c = 2 \text{ V}$$

Region # 3: $(3 \leq t \leq 4)$

In this region v_c is a straight line that does not pass through the origin. The equation of this straight line is

$$v_c = mt + c$$

The slope of this line is -2 and $c = 8$
Therefore

$$v_c = -2t + 8$$

Let us determine i in all these three regions

Region # 1: $(0 \leq t \leq 1)$
As

$$v_c = 2t$$

Therefore

$$i = C\frac{dv_c}{dt} = 20\ A$$

Region # 2: $(1 \leq t \leq 3)$
As

$$v_c = 2\ V$$

Therefore

$$i = C\frac{dv_c}{dt} = 0\ A$$

Region # 3: $(3 \leq t \leq 4)$
As

$$v_c = -2t + 8\ \ V$$

Therefore

$$i = C\frac{dv_c}{dt} = -20\ A$$

Therefore sketch for the current is shown in Figure 1.10.

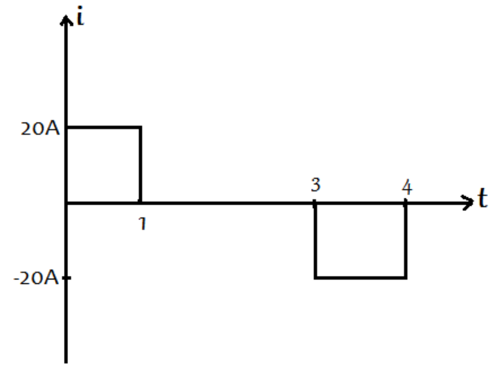

Figure 1.10: Sketch for the Current in the Capacitor

Let us determine p_C in all these three regions:

Region # 1: $(0 \leq t \leq 1)$

As

$$v_c = 2t$$

and

$$i = 20\ A$$

Therefore

$$p_C = v_c \times i = 40t \quad W$$

Region # 2: $(1 \leq t \leq 3)$

As

$$v_c = 2\ V$$

and

$$i = 0\ A$$

Therefore

$$p_C = v_c \times i = 0 \quad W$$

Region # 3: $(3 \leq t \leq 4)$

As

$$v_c = -2t + 8\ \ V$$

and

$$i = -20\ A$$

Therefore

$$p_C = v_c \times i = 40t - 160 \quad W$$

Therefore sketch for the power is shown in Figure 1.11.

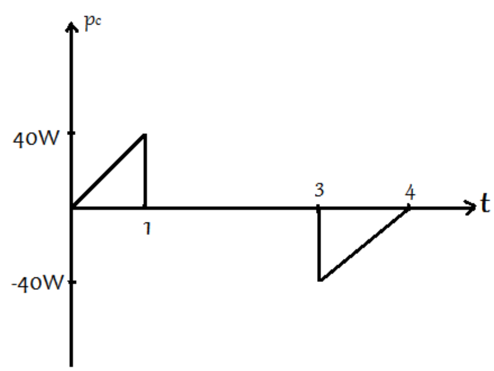

Figure 1.11: Sketch for the Power taken by the Capacitor

D 1.4: Consider the sketch for i as shown in Figure 1.12. Sketch v_L and p_L as a function of time. Inductance of the inductor is 10H.

Figure 1.12: Inductor and Sketch for Current in the Inductor

Solution:

We divide the graph into two regions.

Region # 1: $(0 \leq t \leq 1)$

In this region i is a straight line passing through the origin. Equation of this straight line is

$$i = mt + c$$

Where $m = 2$ is the slope of this line and as the line is passing through the origin, therefore $c = 0$.
So

$$i = 2t$$

Region # 2: $(1 \leq t \leq 2)$

In this region i is a straight line that does not pass through the origin. Equation of this straight line is

$$i = mt + c$$

The slope of this line is -2 and $c = 4$
Therefore

$$i = -2t + 4$$

Let us determine v_L in all these two regions

Region # 1: $(0 \leq t \leq 1)$
As

$$i = 2t$$

Therefore

$$v_L = L\frac{di}{dt} = 20\ V$$

Region # 2: $(1 \leq t \leq 2)$

As

$$i = -2t + 4\ \ V$$

Therefore

$$v_L = L\frac{di}{dt} = -20\ V$$

Therefore sketch for the voltage is shown in Figure 1.13

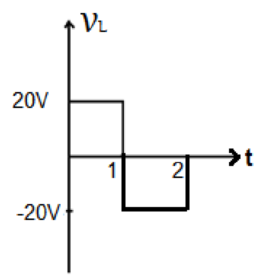

Figure 1.13: Sketch for the Voltage across the Inductor

Let us determine p_L in all these two regions:

Region # 1: $(0 \leq t \leq 1)$

As

$$i = 2t$$

and

$$v_L = 20\ V$$

Therefore

$$p_L = v_L \times i = 40t\ \ W$$

Region # 2: $(1 \leq t \leq 2)$

As

$$i = -2t + 4\ \ V$$

and

$$v_L = -20\ V$$

Therefore

$$p_L = v_L \times i = 40t - 80\ \ W$$

Therefore sketch for the power is shown in Figure 1.14

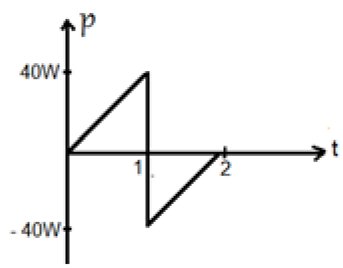

Figure 1.14: Sketch for the Power taken by the Inductor

1-5 Kirchhoff's Voltage Law

This law is known as KVL and is used to find out unknown electrical quantities in a circuit. This law states that *sum of the voltage rises in a loop is always equal to sum of the voltage drops in the loop.* Consider a series combination of three resistors as shown in Figure 1.15.

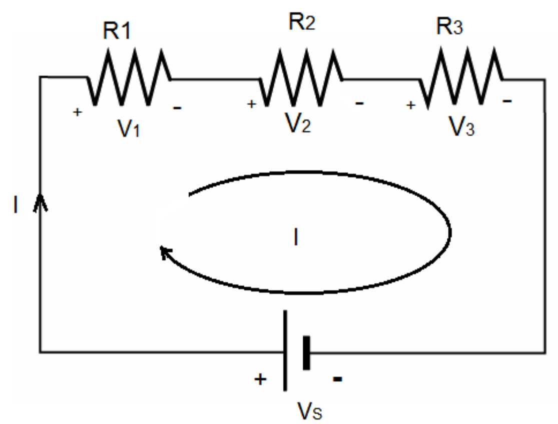

Figure 1.15: Series Circuit

A constant voltage is applied across this series circuit and this voltage source results in a current I that flows in the clockwise direction in the loop. We apply KVL to the given loop

$$V_S = V_1 + V_2 + V_3 \tag{1.36}$$

Equation 1.36 can be rearranged as

$$V_S + (-V_1) + (-V_2) + (-V_3) = 0$$

This equation justifies another statement of KVL. In light of this equation *algebraic sum of all the voltages in a specific direction in a loop is always equal to zero.* Keep it in mind that we place a plus sign with the voltage rise and a minus sign with the voltage drop in this regard.

Voltage drop across R_1 in accordance with ohm's law is given by

$$V_1 = IR_1 \tag{1.37}$$

Voltage drop across R_2 in accordance with ohm's law is given by

$$V_2 = IR_2 \tag{1.38}$$

Voltage drop across R_3 is given by

$$V_3 = IR_3 \tag{1.39}$$

Putting all these values in equation 1.36, we obtain

$$V_S = I(R_1 + R_2 + R_3) \tag{1.40}$$

The current in this series circuit can be found using equation 1.40. Now we replace the series combination of all the three resistors by a single resistor such that the resistance of this single resistor is equal to the total resistance of the series circuit. The equivalent circuit of the above mentioned series circuit is shown in Figure 1.16. Applying KVL to this equivalent circuit, we obtain the following equation

$$V_S = V_T \tag{1.41}$$

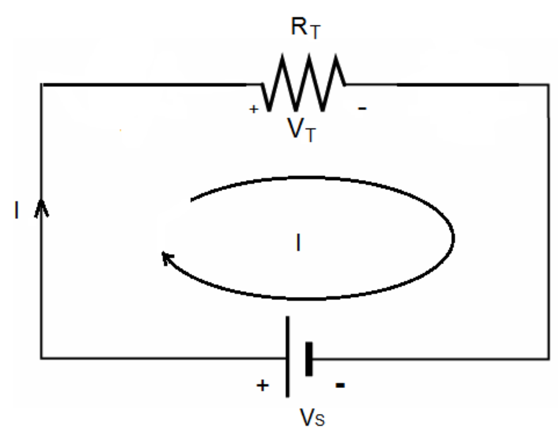

Figure 1.16: Equivalent Circuit

Voltage drop across R_T in accordance with ohm's law is given by

$$V_T = IR_T \tag{1.42}$$

Putting this value in equation 1.41, we obtain

$$V_S = IR_T \tag{1.43}$$

Comparing equation 1.43 with equation 1.40, we obtain total resistance of the series combination of Figure 1.15.

$$R_T = (R_1 + R_2 + R_3) \qquad (1.44)$$

According to the law of conservation of energy power supplied by the voltage source in Figure 1.15 will be equal to the total power taken by the entire circuit.

Power supplied by the voltage source $= P_S = V_s I$

Power consumed by $R_1 = P_1 = I^2R_1$

Power consumed by $R_2 = P_2 = I^2R_2$

Power consumed by $R_3 = P_3 = I^2R_3$

As $P_s = P_T$, therefore

$$P_T = (P_1 + P_2 + P_3) \qquad (1.45)$$

D 1.5: Consider the series circuit as shown in Figure 1.17. The DC voltage source across this series combination is of 10V. Find the current, the voltage drop across each resistor and the power consumed by the entire circuit.

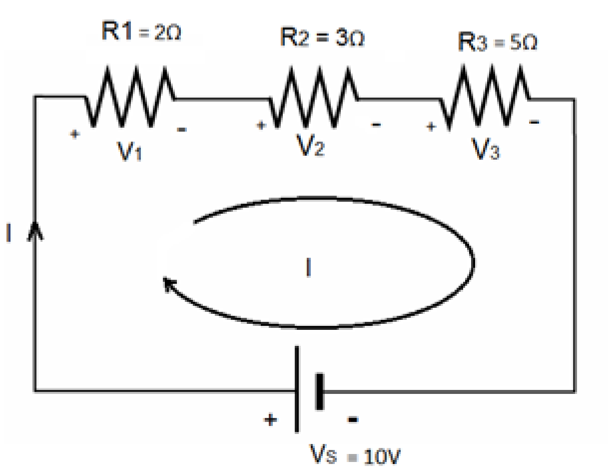

Figure 1.17: Series Circuit

Solution:

Using equation 1.40, we can calculate the current.

$$V_S = I(R_1 + R_2 + R_3)$$

$$I = \frac{V_S}{R_1+R_2+R_3} = \frac{10}{10}$$

$$I = 1A$$

Voltage drop across R_1 is

$$V_1 = IR_1 = 2V$$

Voltage drop across R_2 is

$$V_2 = IR_2 = 3V$$

Voltage drop across R_3 is

$$V_3 = IR_3 = 5V$$

Power supplied by the voltage source $= P_S = V_s\, I = 10 \times 1 = 10W$

1-6 Capacitors in a Series Circuit

Consider a series combination of three capacitors as shown in Figure 1.18. A time varying voltage is applied across this series circuit which results in a time varying current i that flows in the clockwise direction in the loop. We apply kVL to the given loop which states that sum of the voltage rises in this loop will be equal to sum of the voltage drops.

$$v_S = v_1 + v_2 + v_3 \tag{1.46}$$

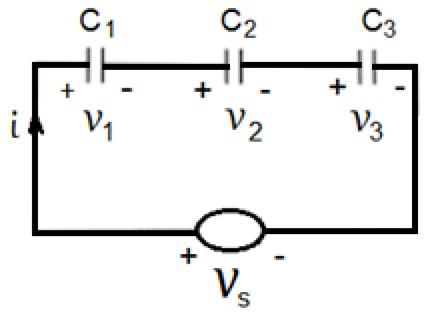

Figure 1.18: Series Circuit of Capacitors

Equation 1.46 can be rearranged as

$$v_S + (-v_1) + (-v_2) + (-v_3) = 0$$

This equation justifies another statement of KVL. In light of this equation KVL states that *algebraic sum of all the voltages in a specific direction in a loop is always equal to zero.* Keep it in mind that we place a plus sign with the voltage rise and a minus sign with the voltage drop in this regard.

Voltage drop across C_1 is

$$v_1 = \frac{1}{C_1}\int idt \tag{1.47}$$

Voltage drop across C_2 is

$$v_2 = \frac{1}{C_2}\int idt \tag{1.48}$$

Voltage drop across C_3 is

$$v_3 = \frac{1}{C_3}\int idt \tag{1.49}$$

Putting all these values in equation 1.46, we obtain

$$v_s = \frac{1}{C_1}\int idt + \frac{1}{C_2}\int idt + \frac{1}{C_3}\int idt \tag{1.50}$$

or

$$v_s = \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right)\int idt \tag{1.51}$$

Now we replace the series combination of all the three capacitors by a single capacitor such that the capacitance of this single capacitor is equal to the total capacitance of the series circuit. The equivalent circuit of the above mentioned series circuit is given in Figure 1.19. Applying KVL to this equivalent circuit, we obtain the following equation.

$$v_S = v_T \tag{1.52}$$

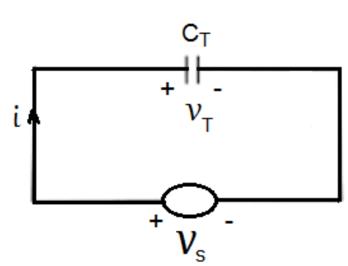

Figure 1.19: Equivalent Circuit

Voltage drop across C_T is

$$v_T = \frac{1}{C_T}\int idt \tag{1.53}$$

Comparing equation 1.53 with equation 1.51, we obtain the total capacitance of the series combination of Figure 1.18.

$$\frac{1}{C_T} = \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right) \tag{1.54}$$

1-7 Kirchhoff's Current Law

This law is known as KCL and is used to find out the unknown electrical quantities in a circuit. This law states that *sum of all the currents flowing towards a node is always equal to sum of all the currents flowing away from the node.* Consider the parallel combination of three resistors as shown in Figure 1.20. A constant voltage is applied across this parallel circuit and this voltage source results in current I_1, I_2 and I_3 as shown in the figure. We apply KCL to the single node of this parallel circuit.

$$I_S = I_1 + I_2 + I_3 \tag{1.55}$$

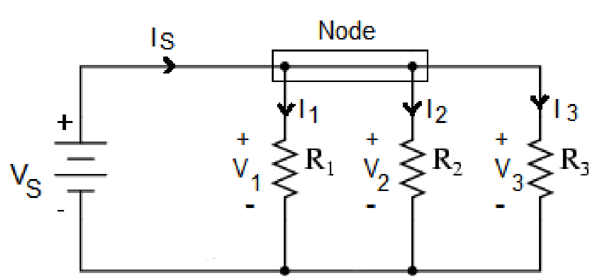

Figure 1.20: Parallel Circuit

It can be easily established that

$$V_S = V_1 = V_2 = V_3 \tag{1.56}$$

Voltage drop across R_1 is

$$V_1 = I_1 R_1$$

Therefore

$$I_1 = \frac{V_S}{R_1} \tag{1.57}$$

Voltage drop across R_2 is

$$V_2 = I_2 R_2$$

Therefore

$$I_2 = \frac{V_S}{R_2} \tag{1.58}$$

Voltage drop across R_3 is

$$V_3 = I_3 R_3$$

Therefore

$$I_3 = \frac{V_S}{R_3} \tag{1.59}$$

Putting all these values of currents in equation 1.55, we obtain

$$I_S = V_S\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right) \tag{1.60}$$

Now we replace the parallel combination of all the three resistors by a single resistor such that the resistance of this single resistor is equal to the total resistance of the parallel circuit. The equivalent circuit of the above mentioned parallel circuit is shown in Figure 1.21. Applying Ohm's law to this equivalent circuit, we obtain the following equation.

$$I_S = \frac{V_S}{R_T} \tag{1.61}$$

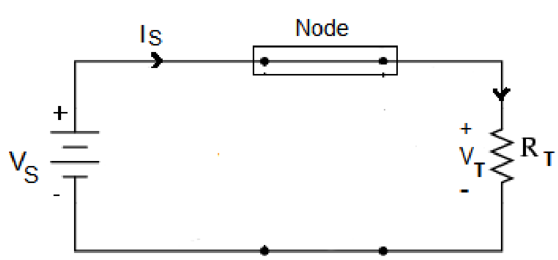

Figure 1.21: Equivalent Circuit

Comparing equation 1.61 with equation 1.60, we obtain the total resistance of the parallel combination of Figure 1.20.

$$\frac{1}{R_T} = \frac{1}{(R_1+R_2+R_3)} \tag{1.62}$$

According to the law of conservation of energy power supplied by the voltage source in Figure 1.20 will be equal to the total power taken by the entire circuit.

Power supplied by the voltage source $= P_S = V_s\, I_S$

Power consumed by $R_1 = P_1 = I_1^2 R_1$

Power consumed by $R_2 = P_2 = I_2^2 R_2$

Power consumed by $R_3 = P_3 = I_3^2 R_3$

As $P_s = P_T$ therefore

$$P_T = (P_1 + P_2 + P_3) \tag{1.63}$$

D 1.6: Consider the parallel circuit as shown in Figure 1.22. The DC voltage across this parallel combination is 16V. Find the currents and the power consumed by the entire circuit.

Solution:

Current through resistor R_1 is

$$I_1 = \frac{V_S}{R_1}$$

$$I_1 = \frac{16}{2} = 8A$$

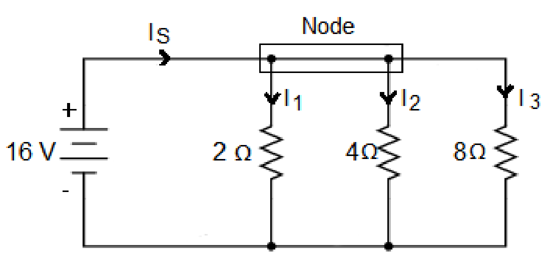

Figure 1.22: Circuit for D # 1.6

Current through resistor R_2 is

$$I_2 = \frac{V_S}{R_2}$$

$$I_2 = \frac{16}{4} = 4A$$

Current through resistor R_3 is

$$I_3 = \frac{V_S}{R_3}$$

$$I_3 = \frac{16}{8} = 2A$$

Power consumed by resistor R_1 is

$$P_1 = I_1^2 R_1 = 64 \times 2 = 128W$$

Power consumed by resistor R_2 is

$$P_2 = I_2^2 R_2 = 16 \times 4 = 64W$$

Power consumed by resistor R_3 is

$$P_3 = I_3^2 R_3 = 4 \times 8 = 32W$$

Total power consumed by the entire circuit is

$$P_t = P_1 + P_2 + P_3 = 224W$$

Current Supplied by the source is

$$I_S = I_1 + I_2 + I_3 = 14A$$

Power Supplied by the source is

$$P_S = V_S I_S = 16 \times 14 = 224W$$

D 1.7: Consider the parallel circuit as shown in Figure 1.23. Find the unknown currents, the node voltage V, the power consumed by the entire circuit and the power supplied by the current sources.

Solution:

Applying KCL to the node

$$I_S = I_1 + I_2 + I_3 \tag{1.64}$$

Sum of the current flowing towards the node is

$$I_S = 8A$$

Current through resistor R_1 is

$$I_1 = \frac{V}{R_1} = 3V$$

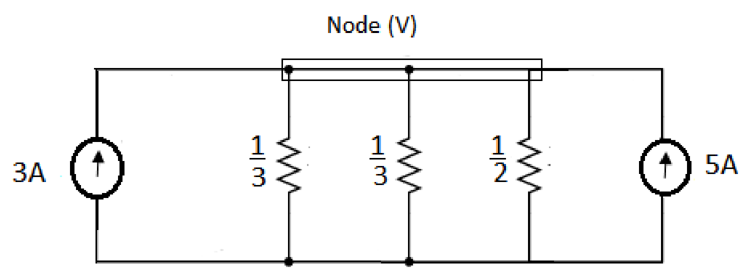

Figure 1.23: Circuit for D # 1.7

Current through resistor R_2 is

$$I_2 = \frac{V}{R_2} = 3V$$

Current through resistor R_3 is

$$I_3 = \frac{V}{R_3} = 2V$$

Putting the values in equation 1.64, we obtain

$$8 = 3V + 3V + 2V$$

$$V = 1 \text{ Volt}$$

So the current I_1 is 3A, the current I_2 is 3A and I_3 is 2A.

Power consumed by resistor R_1 is

$$P_1 = I_1^2 R_1 = 9 \times \frac{1}{3} = 3W$$

Power consumed by resistor R_2 is

$$P_2 = I_2^2 R_2 = 9 \times \frac{1}{3} = 3W$$

Power consumed by resistor R_3 is

$$P_3 = I_3^2 R_3 = 4 \times \frac{1}{2} = 2W$$

Total power consumed by the entire circuit is

$$P_t = P_1 + P_2 + P_3 = 8W$$

Power supplied by the first current source

$$P_{S1} = 1 \times 5 = 5W$$

Power supplied by the second current source

$$P_{S2} = 1 \times 3 = 3W$$

Total power supplied by the two current sources

$$P_S = 8W$$

1-8 Capacitors in a Parallel Circuit

Consider a parallel combination of three capacitors as shown in Figure 1.24. A time varying voltage is applied across this parallel circuit which results in time varying currents i_1 , i_2 and i_3 as shown in the figure.

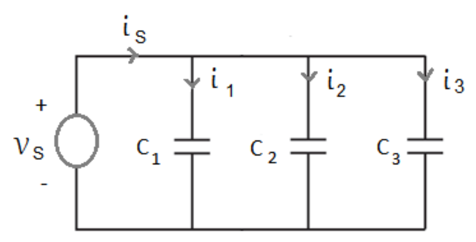

Figure 1.24: Parallel Circuit

We apply KCL to the single node of this parallel circuit.

$$i_S = i_1 + i_2 + i_3 \tag{1.65}$$

This is a parallel circuit and the voltage across each one of the three capacitors is equal to the source voltage. The current in C_1 is given by